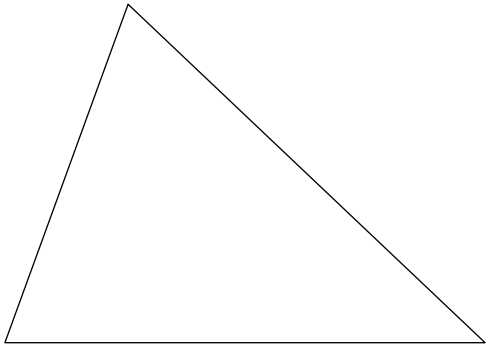
Cyclic Quadrilateral

for the EGMO preparation camp 2025

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Reading

- *Three Lemmas in Geometry - Center of spiral similarity*, Yufei Zhao
- *Cyclic Quadrilaterals — The Big Picture*, Yufei Zhao

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1 Spiral similarity

Lemma 1: Spiral similarity

Let circles o_1, o_2 intersect at A and S, and let two lines through S intersect o_1, o_2 at B, X and C, Y respectively.

Then $\triangle ABC$ is *spiraly similar* to $\triangle AXY$, meaning that they are similar triangles, orientated the same way and they share a common vertex.

$\triangle ABC \backsimeq \triangle AXY$

We then say that the spiral similarity at A sends $B \to X$ and $C \to Y$.

Lemma 2: Spiral pair

Spiral similarities come in pairs - if a spiral similarity at A sends $BC \to XY$ then there also exists a spiral similarity at A that sends $BX \to CY$.

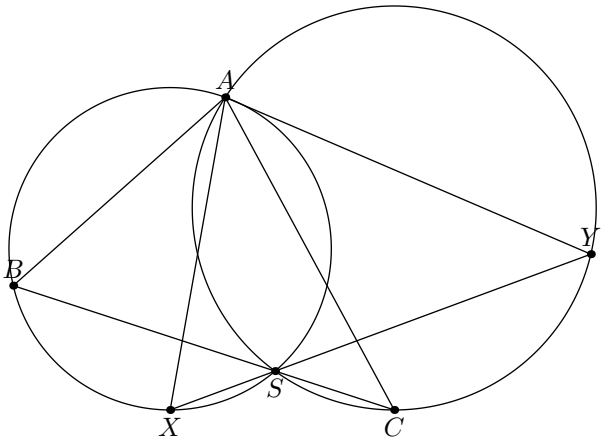


Figure 1: *Spiral*

1.1 Equal lengths

Lemma 3

Let BC be the shortest side of $\triangle ABC$ and D, E be points on AB, AC respectively such that $BD = BC = CE$.

Moreover, let M_{BC} be the midpoint of arc BC containing A. Then $(AM_{BC}DE)$ and $M_{BC}C = M_{BC}E$.

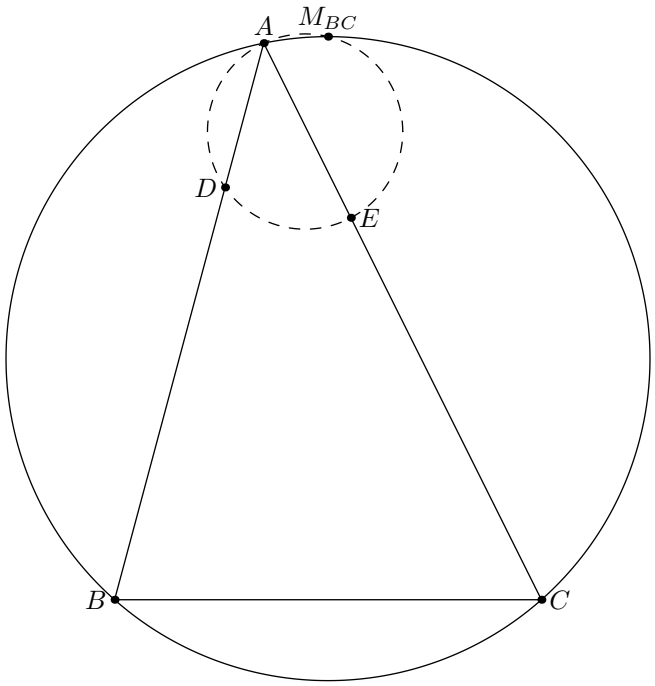


Figure 2: *Lemma 3*

2 Quadrilateral

2.1 Miquel point

Lemma 4

For every quadrilateral $ABCD$, there exists a *Miquel point* that is a center of spiral similarity of $AB \to CD$ and $AC \to BD$.

2.2 Big Picture

Let $ABCD$ be a cyclic quadrilateral with circumcenter O and let $E = AB \cap CD$, $F = AD \cap BC$, $P = AC \cap BD$. Moreover, let M be the *Miquel point* of $(ABCD)$.

Lemma 5: Cyclic Quadrilateral - The Big Picture

- $E - M - F$
- $O - P - M$
- $OM \perp EF$
- $(AOCM)$, $(BODM)$
- MO bisects $\angle CMA$ and $\angle BMD$
- P and M are inverse images of each other wrt $(ABCD)$
- $\triangle PQR$ is *self-polar*
- O is the orthocenter of $\triangle PQR$

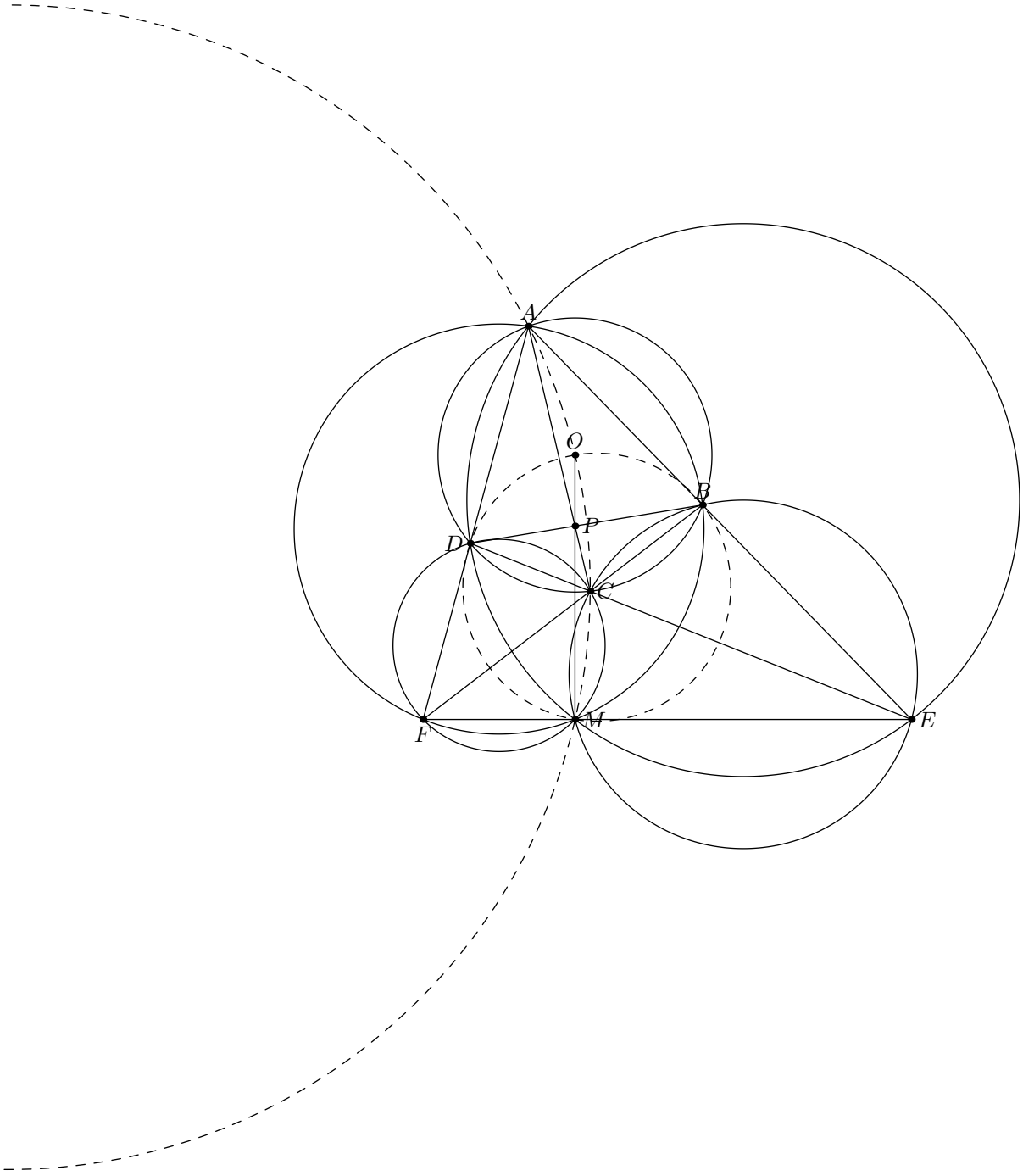


Figure 3: *Lemma 5*

3 Problems

Problem 1. Let $ABCD$ be a quadrilateral. Let diagonals AC and BD meet at P. Let O_1 and O_2 be the circumcenters of $\triangle APD$ and $\triangle BPC$. Let M, N, and O be the midpoints of AC, BD, and O_1O_2. Show that O is the circumcenter of $\triangle MPN$.

Problem 2. Let ABC be a triangle with circumcircle ω. The tangent to ω at B, C meet at T. S lies on the ray BC so that $AS \perp AT$. B_1, C_1 are points on the ray ST (with C_1 between B_1 and SE) such that $B_1T = BT = C_1T$. Prove that $\triangle ABC \sim \triangle AB_1C_1$.

Problem 3. Circles o_1, o_2 with centers O_1, O_2 respectively intersect at points A, B. Line through B cuts o_1, o_2 at C, D. Tangent to o_1, o_2 at C, D respectively cut at E. Line AE intersects $o = (AO_1O_2)$ at $F \neq E$. Prove that the length of segment EF is equal to the length of the diameter of o.

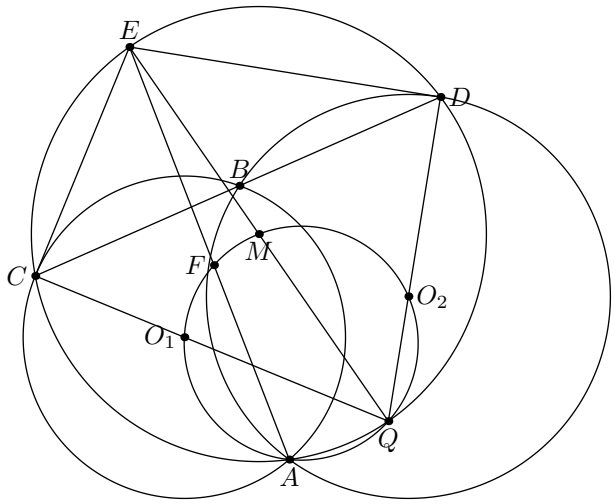


Figure 4: *Problem 3*

Problem 4. Iran TST 2010/5

Circles ω_1 and ω_2 intersect at points P and K. Line XY is tangent to ω_1 at X and ω_2 at Y. Let lines YP and XP meet ω_1 and ω_2 again at B and C, respectively, and let lines BX and CY meet at A. Prove that if the circumcircles of $\triangle ABC$ and $\triangle AXY$ meet again at Q, then $\angle QXA = \angle QKP$.

Problem 5. CAMO 2020/3

Let ABC be a triangle with incircle ω, and let touch BC, CA, AB at D, E, F, respectively. Point M is the midpoint of EF, and T is the point on ω such that DT is a diameter. Line MT meets the line through A parallel to BC at P and ω again at Q. Lines DF and DE intersect line AP at X and Y respectively. Prove that the circumcircles of $\triangle APQ$ and $\triangle DXY$ are tangent.

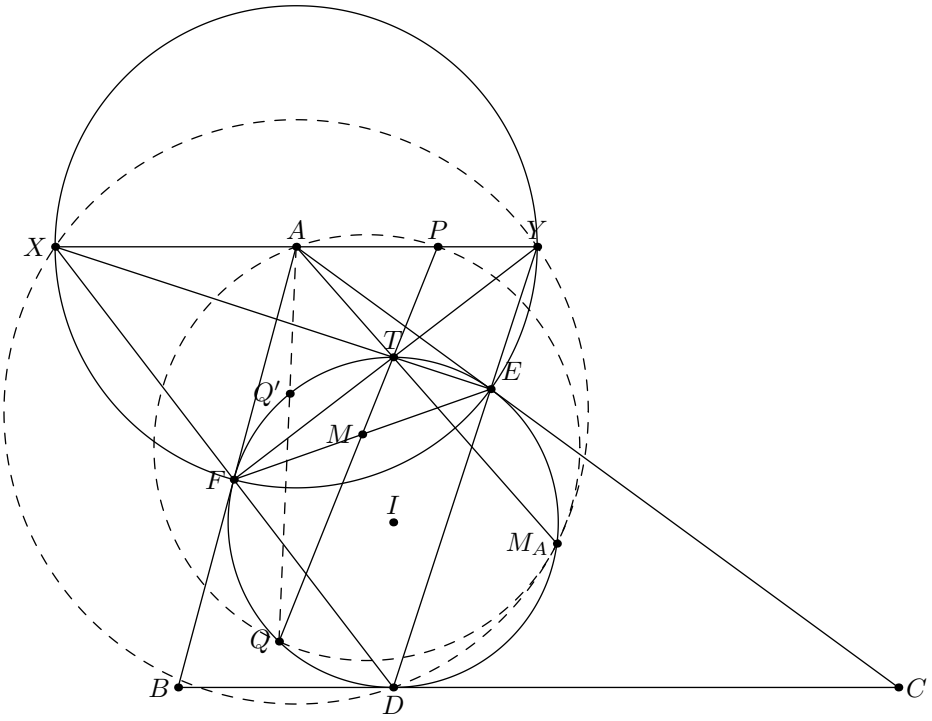


Figure 5: *Problem 5*

3.1 Harder

Problem 6. GOTEEM 2020/5

Let ABC be a triangle and let B_1 and C_1 be variable points on sides BA and CA, respectively, such that $BB_1 = CC_1$. Let $B_2 \neq B_1$ denote the point on (ACB_1) such that BC_1 is parallel to B_1B_2, and let $C_2 \neq C_1$ denote the point on (ABC_1) such that CB_1 is parallel to C_1C_2. Prove that as B_1 and C_1 vary, the circumcircle of AB_2C_2 passes through a fixed point other than A.

Problem 7. Let BC be the shortest side of $\triangle ABC$ and C_0, B_0 be points on AB, AC respectively such that $BC_0 = BC = CB_0$.

Furthermore, let $F = BB_0 \cap CC_0$. Show that incenter I, orthocenter H of $\triangle B_0C_0F$ and M_{BC} midpoint of arc BC containing A lie on one line.

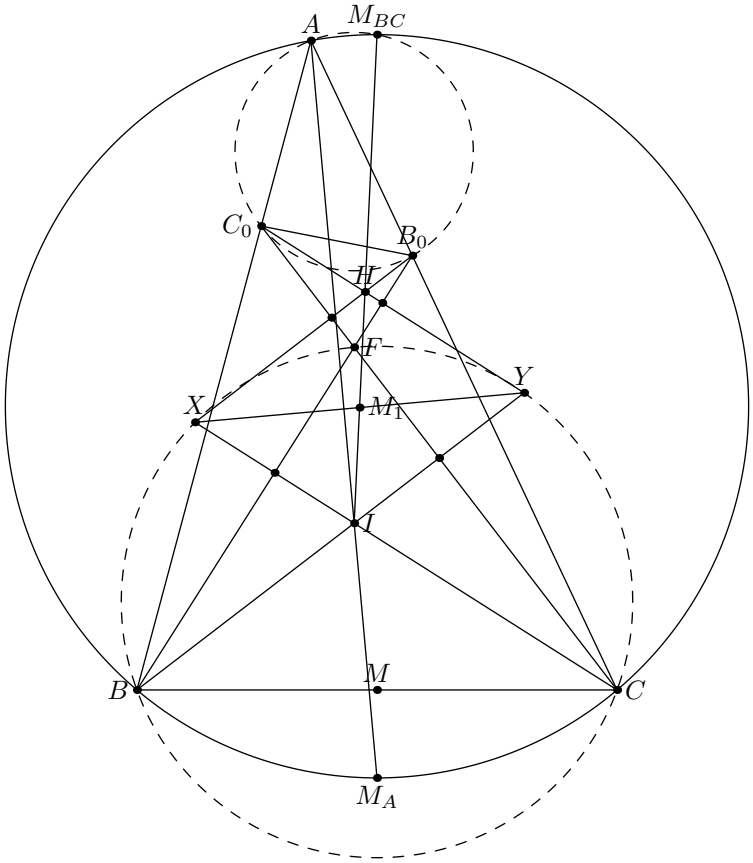


Figure 6: *Problem 7*